

# SPOT-The-Difference: Differentiable Causal Discovery with Skeleton Posterior Guidance

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**Abstract**—Causal discovery attempts to recover cause–effect relationships from observational or interventional data. The problem is difficult because the number of possible graph structures grows super-exponentially with the number of variables, while causal sufficiency is frequently violated by unobserved common causes. Differentiable causal discovery transforms structure learning into a continuous optimization problem, but existing methods may perform poorly when latent confounders are present. This project studies Skeleton Posterior-guided Optimization (SPOT), a two-stage framework that first estimates a posterior distribution over causal skeletons and then uses this posterior to guide differentiable optimization of a maximal ancestral graph. The proposed implementation will be evaluated on the Sachs protein-signaling dataset and on controlled datasets generated with known latent-confounding structures. SPOT will be compared with ABIC and selected constraint-based baselines. The evaluation will measure skeleton recovery, directed-edge recovery where identifiable, structural Hamming distance, precision, recall, F1 score, false-positive rate, and computational cost. The primary research question is whether posterior guidance reduces the search space and improves graph recovery under unmeasured confounding relative to unguided differentiable optimization.

**Index Terms**—causal discovery, differentiable structure learning, latent confounding, ADMG, MAG, skeleton posterior, SPOT, Sachs dataset

## I. PROBLEM DEFINITION AND MOTIVATION

### A. Causal Discovery

A dataset may contain measurements of several variables without providing the causal structure that generated those measurements. A probabilistic model can describe statistical dependence, but causal reasoning requires additional structure that distinguishes a cause from an effect. A causal graph represents this structure and can be used to reason about interventions, counterfactuals, and the effect of changing one or more variables [1].

In this project, the observed data are represented as

$$\mathcal{D} = \{x^{(i)}\}_{i=1}^n, \quad x^{(i)} \in \mathbb{R}^d,$$

where  $n$  is the number of samples and  $d$  is the number of observed variables. The objective is to estimate a graph  $G$  that explains the distribution of the observations and, where possible, identifies the direction of causal relationships.

When all relevant variables are observed and the system is acyclic, a directed acyclic graph (DAG) is a natural representation. However, biological systems commonly contain variables that are not measured. An unobserved variable that causes two

or more observed variables creates latent confounding. Treating such dependence as a direct observed-to-observed causal relationship may result in incorrect edges and orientations.

### B. Differentiable Causal Discovery

Traditional causal-discovery algorithms search over discrete graph structures. This search is computationally expensive, particularly as the number of variables increases. Differentiable methods instead represent candidate edges with continuous parameters and optimize a data-dependent objective using gradient-based procedures.

For a weighted adjacency matrix  $A$ , the DAG constraint used by NOTEARS-style methods can be written as

$$h(A) = \text{tr}(e^{A \circ A}) - d = 0,$$

where  $\circ$  denotes the Hadamard product. The constraint is zero if and only if the graph represented by  $A$  is acyclic under the corresponding assumptions [2].

Although this formulation avoids explicit combinatorial search, it does not by itself solve the problem of latent confounding. A method that assumes a DAG over observed variables can incorrectly explain confounding through directed edges among the observed variables.

### C. Latent Confounders and SPOT

When latent variables are present, the observed distribution is more appropriately represented by an acyclic directed mixed graph (ADMG) or a maximal ancestral graph (MAG). Directed edges represent causal relationships among observed variables, while bidirected edges represent the presence of latent common causes.

The proposed project investigates SPOT, or Skeleton Posterior-guided Optimization. SPOT separates the problem into two stages:

- 1) estimate a posterior distribution over possible causal skeletons; and
- 2) use samples or probabilities from this posterior to guide differentiable optimization of the graph structure.

A skeleton records adjacency without committing to edge orientation. Consequently, it can preserve information about which variables are connected while postponing decisions about whether a relationship is directed or confounded. This reduces the optimization search space and allows uncertainty in adjacency to be represented explicitly [4].

#### D. Core Problem Statement

The project seeks to implement and evaluate a differentiable causal discovery procedure based on SPOT. The system will be tested under observed-variable settings and under controlled variable-omission settings in which variables that act as confounders are deliberately removed from the data.

The main research question is:

Can skeleton-posterior guidance improve causal-graph recovery and computational efficiency when differentiable causal discovery is performed in the presence of latent confounders?

The working hypothesis is that SPOT will achieve higher skeleton-level F1 and lower structural Hamming distance than an unguided differentiable baseline, particularly as the proportion of omitted variables increases.

#### II. MATHEMATICAL QUESTION

Let  $X = (X_1, \dots, X_d)$  be the observed variables and let  $Z$  denote unobserved variables. A structural causal model can be written as

$$X_j = f_j(\text{Pa}_j, Z_j, \epsilon_j), \quad j = 1, \dots, d,$$

where  $\text{Pa}_j$  denotes the observed parents of  $X_j$  and  $\epsilon_j$  denotes exogenous noise.

The observed distribution is

$$p(x) = \int p(x, z) dz.$$

Because  $Z$  is unavailable, multiple causal graphs may imply the same observational distribution. The task is therefore not always to recover one uniquely identifiable DAG. Instead, the goal is to recover the equivalence class or a suitable mixed graph that encodes the causal relations identifiable from the available data.

Let  $S$  denote a binary skeleton adjacency matrix, where

$$S_{ij} = \begin{cases} 1, & \text{if } X_i \text{ and } X_j \text{ are adjacent,} \\ 0, & \text{otherwise.} \end{cases}$$

SPOT estimates

$$q_\phi(S \mid X),$$

where  $\phi$  denotes the parameters of the skeleton-posterior model. The resulting posterior is used to guide optimization of a mixed graph parameterized by directed-edge weights  $A$  and bidirected-edge weights  $B$ .

A general objective can be written as

$$\min_{A, B, \theta} \mathcal{L}_{\text{fit}}(A, B, \theta; \mathcal{D}) + \lambda_A \mathcal{R}_A(A) + \lambda_B \mathcal{R}_B(B) + \lambda_S \mathcal{L}_{\text{SPOT}}(A, B; q_\phi)$$

subject to graph-validity constraints. Here,  $\theta$  denotes model parameters,  $\mathcal{L}_{\text{fit}}$  is a likelihood or reconstruction loss,  $\mathcal{R}_A$  and  $\mathcal{R}_B$  regularize the graph, and  $\mathcal{L}_{\text{SPOT}}$  encourages the optimized graph to remain compatible with likely skeletons.

The project will investigate the following measurable hypotheses:

$$H_1 : \text{F1}_{\text{SPOT}} > \text{F1}_{\text{ABIC}}, \quad (1)$$

$$H_2 : \text{SHD}_{\text{SPOT}} < \text{SHD}_{\text{ABIC}}, \quad (2)$$

$$H_3 : \text{FPR}_{\text{SPOT}} < \text{FPR}_{\text{ABIC}}, \quad (3)$$

$$H_4 : \text{the advantage of SPOT increases with confounding severity.} \quad (4)$$

These hypotheses will be tested on both synthetic data with known ground truth and the Sachs dataset.

#### III. ASSUMPTIONS MADE AND CONSTRAINTS

The following assumptions define the initial scope of the project.

- The samples are treated as independent and identically distributed within each experimental setting.
- The observed variables are measured without substantial measurement error after preprocessing.
- The underlying causal system can be represented adequately by an acyclic mixed graph for the primary experiments.
- The observed distribution contains sufficient statistical information for estimating at least part of the skeleton.
- The conditional-independence and distributional assumptions of each baseline are applied consistently across methods.
- Hyperparameters are selected using only the training or validation portion of the data.
- The Sachs consensus graph is used as an evaluation reference, while recognizing that biological consensus graphs are not perfect ground truth.

The main constraints are the small dimensionality of the Sachs dataset, the limited number of observational samples, possible violations of linearity and Gaussianity, and incomplete identifiability of edge orientations. In particular, a correct evaluation must not penalize a method for failing to orient an edge that cannot be identified from the available data.

The project therefore prioritizes skeleton-level evaluation on Sachs. Directed-edge metrics will be reported separately and only where the reference graph and evaluation protocol justify them.

#### IV. DATASET AND BASELINE ALGORITHMS

##### A. Sachs Protein-Signaling Dataset

The primary real-world dataset is the Sachs protein-signaling dataset. It contains measurements of 11 proteins and phospholipids involved in cell-signaling pathways. The observational subset contains 853 samples, while the broader dataset includes measurements collected under several experimental conditions [5].

The variables include:

|      |     |      |      |
|------|-----|------|------|
| Raf  | Mek | Plcg | PIP2 |
| PIP3 | Erk | Akt  | PKC  |
| JNK  | PKA | p38  |      |

The main experiment will use the observational subset to ensure that the proposed method is evaluated in the intended

observational causal- discovery setting. If the implementation supports intervention labels, the interventional observations will be used in a separate supplementary experiment rather than mixed silently with observational data.

The Sachs data will be standardized using a documented preprocessing pipeline. The preprocessing stage will include removal of metadata columns from the feature matrix, handling of missing values if present, and feature scaling. No preprocessing operation will use the reference graph.

### B. Controlled Latent-Confounding Experiments

Because the actual latent variables in Sachs are not directly observed, controlled experiments will supplement the real-world evaluation. A known graph will first be sampled. Latent variables will then be added as common causes of selected observed-variable pairs. The latent variables will be removed before training.

The confounding level will be controlled using:

- 1) the number of latent variables;
- 2) the fraction of observed variables affected by latent causes;
- 3) the strength of the latent-to-observed coefficients; and
- 4) the number of observed variables omitted from the dataset.

This protocol makes it possible to measure recovery against a known ground-truth ADMG or MAG and to determine whether performance changes systematically with confounding severity.

### C. ABIC Baseline

ABIC is the principal differentiable baseline. It learns a differentiable mixed-graph representation through continuous optimization under assumptions associated with the underlying structural-equation model. The baseline will be run with the same preprocessing, train-validation splits, initialization budget, and stopping criteria as SPOT.

The comparison will distinguish between:

- vanilla ABIC without skeleton guidance; and
- ABIC augmented with the proposed skeleton-posterior guidance.

This ablation is necessary because an improvement over ABIC should be attributable to the skeleton posterior rather than to unrelated implementation differences.

### D. Constraint-Based Baselines

FCI or RFCI will be included where a compatible implementation and evaluation representation are available. These methods are designed to handle latent and selection variables through conditional-independence testing and return a partial ancestral graph or a related equivalence- class representation.

Constraint-based methods provide a complementary comparison to the continuous optimization methods. They may be more robust to some model- misspecification settings but can become expensive as the conditioning sets grow and can be sensitive to finite-sample conditional-independence tests.

TABLE I  
METHODS AND INTENDED COMPARISONS

| Method       | Purpose                                                                                |
|--------------|----------------------------------------------------------------------------------------|
| SPOT         | Proposed skeleton-posterior-guided differentiable discovery method.                    |
| ABIC         | Unguided differentiable discovery baseline for latent confounding.                     |
| FCI/RFCI     | Constraint-based baselines that represent latent-confounding uncertainty.              |
| Ablated SPOT | Tests the effect of removing posterior guidance or replacing it with a point estimate. |

## V. STATE OF THE ART AND RELATED WORK

### A. Structural Causal Models

Pearl’s structural causal-model framework separates statistical association from causal intervention. A structural model consists of variables, functions, and exogenous disturbances. The graph induced by the model determines causal paths, while intervention is represented by replacing one or more structural equations [1].

The framework also provides graphical criteria such as d-separation for reasoning about conditional independence. Under latent variables, the observed-variable graph does not generally remain a DAG. Bidirected edges in an ADMG represent unobserved common causes and preserve information that would be lost by forcing the model into a DAG.

### B. Continuous DAG Optimization

NOTEARS demonstrated that acyclicity can be expressed as a smooth equality constraint, enabling gradient-based optimization over a continuous adjacency matrix [2]. This result motivated a family of differentiable structure-learning algorithms.

The limitation for this project is that the original formulation is primarily designed for DAGs under observed-variable assumptions. When a latent confounder affects two observed variables, a directed-only model may create spurious orientations or additional directed paths.

### C. Differentiable Discovery with Unmeasured Confounding

Bhattacharya *et al.* developed differentiable constraints for learning ADMGs under unmeasured confounding. Their formulation includes ancestral, arid, and bow-free classes of ADMGs and casts graph recovery as a continuous optimization problem [3].

This work provides the mathematical foundation for differentiable latent-confounding discovery. Its limitation is that optimization still operates over a large graph space. The project therefore investigates whether estimating a skeleton posterior before graph optimization can make the search more accurate and scalable.

### D. SPOT

SPOT addresses the latent-confounding problem through a two-phase procedure. First, it uses amortized inference and supervised causal learning to estimate a posterior over

TABLE II  
SUMMARY OF RELATED WORK AND PROJECT POSITIONING

| Work     | Main contribution                                 | Relation to this project                                       |
|----------|---------------------------------------------------|----------------------------------------------------------------|
| Pearl    | SCMs, interventions, graphical causal reasoning   | Provides the causal semantics and identifiability perspective. |
| NOTEARS  | Smooth DAG                                        | Provides the basis for continuous structure optimization.      |
| ABIC     | Differentiable discovery under latent confounding | Principal unguided baseline.                                   |
| SPOT     | Skeleton posterior-guided optimization            | Proposed method under evaluation.                              |
| FCI/RFCI | Constraint-based discovery with latent variables  | Complementary non-differentiable baselines.                    |

skeletons. Second, it samples or uses posterior information to guide differentiable optimization of MAGs.

The central distinction is between posterior estimation and point-estimated adjacency selection. Rather than committing immediately to one skeleton, the method retains uncertainty about possible adjacencies. This is useful when finite samples, latent variables, or non-Gaussian observations make individual edge decisions unreliable.

The SPOT paper reports experiments on synthetic graphs and real-world data, including Sachs. Its reported Sachs evaluation is performed at the skeleton level because edge endpoints are not fully identifiable. This motivates the evaluation design adopted in this report.

#### E. Project Positioning

The project contribution is not a new causal semantics or a claim that all Sachs relationships are identifiable. The contribution is an implementation and empirical evaluation of posterior-guided differentiable structure learning, with particular attention to latent-confounder stress tests and reproducible evaluation.

### VI. MAPPING TO KEY AREAS

#### A. Representation

The representation layer defines the variables, graph class, and parameters used by the learning algorithm. The project will represent the learned structure using:

- an observed-variable skeleton;
- a directed adjacency matrix for candidate causal edges;
- a bidirected adjacency matrix for candidate confounding edges;
- a partial ancestral representation when orientations are not identifiable; and
- posterior edge probabilities produced by the skeleton model.

The representation must distinguish absence of an edge from uncertainty about an edge. A low posterior probability is not equivalent to a proven absence until a decision threshold is applied.

TABLE III  
PROJECT EVALUATION KPIs

| Criterion                   | Target or assessment method                                                      |
|-----------------------------|----------------------------------------------------------------------------------|
| Skeleton F1                 | SPOT should improve over ABIC on Sachs and latent-confounding simulations.       |
| Skeleton precision          | Measure the fraction of predicted adjacencies that are correct.                  |
| Skeleton recall             | Measure the fraction of reference adjacencies recovered.                         |
| Structural Hamming distance | Report additions, deletions, and orientation errors separately where possible.   |
| False-positive rate         | Assess spurious adjacencies caused by latent confounding.                        |
| Confounding robustness      | Evaluate performance across increasing latent-variable strength.                 |
| Runtime                     | Compare training and inference time under equal hardware budgets.                |
| Posterior calibration       | Measure whether predicted adjacency probabilities reflect empirical correctness. |
| Reproducibility             | Report mean and standard deviation over at least five random seeds.              |

#### B. Inference

Inference is used in two places. First, the skeleton-posterior model estimates

$$q_\phi(S \mid X).$$

Second, the differentiable optimizer estimates graph parameters and model parameters conditional on the posterior guidance.

The implementation will record posterior calibration in addition to thresholded graph quality. Calibration can be evaluated using Brier score or reliability curves for the binary adjacency events.

#### C. Learning

The learning pipeline consists of:

- 1) data preprocessing and split generation;
- 2) skeleton-posterior estimation;
- 3) posterior-guided mixed-graph optimization;
- 4) thresholding or graph decoding;
- 5) equivalence-aware evaluation; and
- 6) repeated-seed aggregation.

All methods will use matching computational budgets whenever possible. The random seeds, hyperparameters, stopping rules, and hardware configuration will be logged.

### VII. INITIAL KEY PERFORMANCE INDICATORS

The KPIs are divided into accuracy, robustness, scalability, and reproducibility.

For a predicted skeleton  $\hat{S}$  and reference skeleton  $S^*$ , skeleton precision, recall, and F1 are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$

and

$$F_1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Structural Hamming distance will be reported as

$$\text{SHD} = \text{deletions} + \text{additions} + \text{orientation errors}.$$

For Sachs, the primary score will be skeleton SHD because several directed endpoints may not be identifiable.

The target for the initial milestone is a complete and reproducible implementation in which:

- SPOT and ABIC run on identical data splits;
- the Sachs preprocessing pipeline is fixed and documented;
- synthetic latent-confounding experiments have known ground truth;
- all primary metrics are computed automatically; and
- results are reported over multiple random seeds.

The project will not impose an arbitrary universal accuracy threshold on the Sachs dataset. Instead, performance will be interpreted relative to ABIC, the selected constraint-based methods, and the reference graph, because the Sachs graph is a biological consensus rather than a perfect oracle.

## VIII. EXPERIMENTAL PLAN

The experimental evaluation consists of two related benchmark settings using the Sachs protein-signaling dataset. First, the proposed SPOT implementation will be compared with ABIC using the complete dataset. Second, the same comparison will be repeated after selected variables have been removed from the observed data.

### A. Experiment 1: Complete Sachs Dataset

The first experiment evaluates SPOT and ABIC using the complete set of variables selected for the Sachs benchmark. Both methods will receive the same preprocessed data, variable ordering, data partitions, random-seed settings, and computational resources.

The methods will be compared using:

- skeleton precision;
- skeleton recall;
- skeleton F1 score;
- structural Hamming distance;
- true-positive rate;
- false-positive rate; and
- execution time.

The primary objective is to determine whether skeleton-posterior guidance enables SPOT to recover the Sachs adjacency structure more accurately than ABIC. Skeleton-level evaluation is used as the primary evaluation because some edge orientations may not be identifiable from the available observations.

### B. Experiment 2: Sachs with Variables Removed

The second experiment evaluates robustness to incomplete observation. Predefined variables will be removed from the Sachs dataset before either method is trained. The removed variables will not be provided to the models and will therefore act as unobserved variables from the perspective of the remaining observed variables.

The experiment will follow these steps:

- 1) Select the variables to be removed before running the final experiments.
- 2) Remove the selected variables from the input data.
- 3) Train SPOT and ABIC using the remaining variables.
- 4) Construct the corresponding reference skeleton over the remaining variables.
- 5) Compute the same graph-recovery and runtime metrics used in the complete-data experiment.

The selected removal configurations will include variables that are connected to multiple other variables in the reference graph. Removing such variables provides a meaningful test of whether the methods remain effective when unobserved variables can induce dependence among the remaining observed variables.

The principal analysis will compare the performance decrease between the complete-data and variable-removed settings. Robustness will be assessed using the change in skeleton F1, structural Hamming distance, precision, recall, and false-positive rate.

### C. Common Evaluation Protocol

SPOT and ABIC will use the same preprocessing pipeline, variable ordering, data splits, stopping criteria, and computational environment. Each method will be executed using the same random seeds. Results will be reported as the mean and standard deviation over the selected runs.

For a predicted skeleton  $\hat{S}$  and a reference skeleton  $S^*$ , precision, recall, and F1 score are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$

and

$$F_1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Structural Hamming distance will be computed primarily at the skeleton level. Where an orientation is not identifiable, it will not be treated as an ordinary orientation error. Directed-edge results, if reported, will be presented separately from the primary skeleton metrics.

The final evaluation will contain two main result tables. The first will report the comparison between SPOT and ABIC on the complete Sachs dataset. The second will report the same comparison after variables have been removed.

### D. Expected Comparison

The evaluation will test the following hypotheses:

- SPOT will achieve a higher or comparable skeleton F1 score than ABIC on the complete Sachs dataset.
- SPOT will experience a smaller reduction in skeleton F1 after variables are removed.
- SPOT will produce fewer false-positive adjacencies under incomplete observation.
- Any accuracy improvement will be assessed together with execution time to determine its computational cost.

These statements define the experimental hypotheses and will not be treated as guaranteed results. The report will distinguish between expected behavior and measured performance.

TABLE IV  
TIMELINE AND TEAM RESPONSIBILITIES

| Member        | Assigned responsibility                                                                                                                                                                                                                                                                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anant Narayan | Lead the literature review covering causal discovery, differentiable structure learning, latent confounding, MAGs, ADMGs, ABIC, SPOT, and Pearl’s causal framework. Implement the data-preparation and experiment-control module, including Sachs loading, preprocessing, variable-removal configurations, data splits, and reproducible random-seed management. |
| Ankit Kumar   | Implement the ABIC baseline module. Configure ABIC for the complete Sachs dataset and the variable-removed settings. Develop the baseline training interface, parameter configuration, graph export, and runtime logging.                                                                                                                                        |
| Milan Shah    | Implement the SPOT module. Develop the skeleton-posterior estimation, posterior-guided optimization, graph decoding, and graph-export components required for the benchmark.                                                                                                                                                                                     |
| All members   | Integrate the three modules, use a common evaluation interface, review each other’s code, reproduce all benchmark runs, analyze the results, and contribute to the final report and presentation.                                                                                                                                                                |

## IX. TIMELINE AND TEAM RESPONSIBILITIES

The implementation is divided into three modules of comparable scope. Each team member is responsible for one module, while integration, testing, and final evaluation are shared responsibilities.

The three individual implementation assignments are:

- 1) Anant: literature review, data preparation, experiment control and ABIC baseline implementation;
- 2) Ankit: data preparation, ABIC baseline implementation and literature review
- 3) Milan: SPOT implementation and experiment control

The modules are intended to represent equal portions of the implementation effort. As we move through the project we may alter or shift responsibilities as needed.

Each member will review at least one other member’s module. The final reported results will be generated through the shared evaluation pipeline rather than through separate, independently implemented metric scripts.

## X. EXPECTED CONTRIBUTIONS AND LIMITATIONS

The expected contributions are:

- 1) an implementation of a skeleton-posterior-guided differentiable causal-discovery pipeline;
- 2) a reproducible comparison with ABIC and selected constraint-based methods;
- 3) a controlled study of performance under increasing latent confounding; and
- 4) an evaluation on the Sachs protein-signaling dataset using skeleton-aware metrics.

The project has several limitations. First, the Sachs reference graph is not a complete ground-truth causal graph. Second, observational data alone may not identify all edge directions.

TABLE V  
MILESTONE ROADMAP

| Phase       | Focus                                    | Expected output                                                                                               |
|-------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Milestone 1 | Problem definition and literature review | Causal-discovery formulation, SPOT and ABIC review, Sachs data protocol, assumptions, and evaluation metrics. |
| Milestone 2 | Individual implementation                | Data-preparation module, ABIC baseline, SPOT pipeline, and common graph-output format.                        |
| Milestone 3 | Pipeline integration                     | Complete Sachs benchmark, variable-removal configurations, and shared evaluation scripts.                     |
| Final phase | Evaluation and reporting                 | Benchmark tables, runtime measurements, result analysis, limitations, final report, and presentation.         |

Third, the assumptions behind continuous optimization may not hold for every protein-signaling relationship. Fourth, the learned skeleton posterior can itself be miscalibrated when the training distribution differs substantially from the evaluation data. Finally, improved graph-recovery metrics do not automatically establish that every learned edge represents a biologically valid mechanism.

These limitations will be addressed by reporting uncertainty, using synthetic data with known ground truth, separating skeleton and orientation metrics, and including ablation experiments.

## XI. CONCLUSIONS

This report defines a project for differentiable causal discovery in the presence of latent confounders. The central challenge is that differentiable optimization makes graph learning computationally tractable but does not by itself resolve the ambiguity created by unobserved common causes. SPOT addresses this challenge by estimating a posterior over causal skeletons and using that posterior to guide optimization of a mixed causal graph.

The proposed evaluation uses synthetic latent-confounding experiments and the Sachs protein-signaling dataset. ABIC provides the principal unguided differentiable baseline, while FCI or RFCI provide a constraint-based comparison where feasible. Skeleton-level metrics are the primary evaluation criteria on Sachs because causal endpoint orientation is not uniformly identifiable.

The ultimate deliverable is a reproducible experimental pipeline covering preprocessing, posterior estimation, graph optimization, baseline comparison, ablation, and evaluation. Later milestones will determine whether posterior guidance improves accuracy, robustness, and computational behavior over unguided differentiable causal discovery.

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